

Metropolitan General Hospital - Final Medical Summary

PATIENT: COX, Harold

MRN: AML067

DOB: April 21, 1946

DATE OF ADMISSION: April 10, 2023

DATE OF DEATH: May 2, 2023 at 08:15

SUMMARY OF HOSPITAL COURSE:

Mr. Cox was a 77-year-old gentleman with a history of adverse-risk Acute Myeloid Leukemia who was admitted for management of complications related to his terminal illness.

Background:

Mr. Cox was diagnosed with de novo AML in July 2022. His diagnostic bone marrow biopsy was significant for a **KMT2A (MLL) gene rearrangement**, a known high-risk cytogenetic feature. Given his age but otherwise excellent functional status (ECOG 1), he was considered a candidate for intensive chemotherapy. The specific regimen chosen was **CPX-351 (Vyxeos)**, a liposomal formulation of daunorubicin and cytarabine, which has shown efficacy in patients with high-risk features.

He began his induction therapy on August 5, 2022. He tolerated the regimen relatively well and achieved a complete remission. He subsequently received two cycles of consolidation therapy with CPX-351, with the final cycle completed in December 2022.

In early April 2023, he presented to the oncology clinic with rapidly progressing fatigue, bleeding gums, and petechiae. His bloodwork confirmed a relapse of his leukemia, with 55% blasts in the peripheral blood.

Final Hospitalization:

Mr. Cox was admitted on April 10, 2023, for management of symptomatic pancytopenia and to discuss any further options. Upon admission, he was transfusion-dependent and suffering from significant bone pain and profound weakness.

A multidisciplinary meeting was held with Mr. Cox, his wife, and his son. We discussed the extremely poor prognosis of relapsed KMT2A-rearranged AML, particularly in an older adult. We reviewed potential salvage therapies, including clinical trials and other chemotherapy regimens. However, we explained that the likelihood of achieving a meaningful or durable second remission was less than 10%, and the toxicity of such treatments would be substantial.

Mr. Cox, who was clear and lucid, expressed that he had "fought a good fight" but did not wish to spend his remaining time in the hospital receiving toxic and likely futile chemotherapy. He made a definitive choice to forego further anti-leukemic therapy and to focus on comfort.

The Palliative Care service was consulted to assist with symptom management. His care plan was refocused entirely on quality of life. He was started on a continuous infusion of hydromorphone (Dilaudid) via a Patient-Controlled Analgesia (PCA) pump, which provided excellent control of his bone pain. He received platelet transfusions as needed to manage oral bleeding.

Over the next three weeks, he experienced a gradual, peaceful decline. His family was a constant presence at his bedside. In his final days, he became progressively less responsive and passed away quietly on the morning of May 2, 2023.

CAUSE OF DEATH:

- **Primary:** Complications of Refractory Acute Myeloid Leukemia

No autopsy was requested.